

On the Mathematical Structure of the T0 Theory:

Why Numerical Ratios Must Not Be Directly Simplified

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Original

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On the Mathematical Structure of the T0-Theory: Why Numerical Ratios Must Not Be Directly Simplified

Note on versions. The German and English versions of this document differ in structure and extent; earlier revisions shortened, extended or changed parts of both. Where they differ, the more extensive German version applies; sections contained only in this English version count as supplements.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

Introduction

In theoretical physics, the question often arises as to which mathematical operations are legitimate and which are not. A particularly interesting problem occurs in the T0-theory, where seemingly simple numerical ratios such as $\frac{2}{3}$ and $\frac{8}{5}$ possess a deeper structural significance that prohibits direct simplification.

The Fundamental Problem

The T0-theory postulates two equivalent representations for the lepton masses:

$$\text{Simple Form: } m_e = \frac{2}{3} \cdot \xi^{5/2}, \quad m_\mu = \frac{8}{5} \cdot \xi^2$$

$$\text{Extended Form: } m_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \xi^{5/2}, \quad m_\mu = \frac{9}{4\pi\alpha} \cdot \xi^2$$

At first glance, one might assume that the fractions $\frac{2}{3}$ and $\frac{8}{5}$ are simple rational numbers that could be simplified or reduced. However, this assumption would be incorrect.

Why Direct Simplification Is Not Allowed

Equating both representations leads to:

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \quad \frac{8}{5} = \frac{9}{4\pi\alpha}$$

These equations show that the seemingly simple fractions are, in fact, composite expressions containing the constants (π , α) and geometric factors ($\sqrt{3}$).

Mathematical and Physical Consequences

1. **Structure Preservation:** Direct simplification would destroy the underlying geometric and physical structure.
2. **Information Loss:** The fractions encode information about spacetime geometry and electromagnetic coupling.
3. **Equivalence Principle:** Both representations are mathematically equivalent, but the extended form reveals the physical origin.

.1 Circular Relationships and Fundamental Constants

In the T0-theory, seemingly circular relationships arise, which express how the constants are linked in the T0 framework:

$$\begin{aligned}\alpha &= f(\xi) \\ \xi &= g(\alpha)\end{aligned}$$

Here f denotes a general dependence, not a frequency and not the base winding number $f = 1/\xi$.

This mutual dependence leads to an apparent chicken-and-egg problem: Which comes first, α or ξ ?

Resolution of the Circularity Problem

The solution lies in the realization that both constants are expressions of an underlying geometric structure:

α and ξ are not independent of each other but are emergent properties of the fractal spacetime geometry.

The apparent circularity dissolves when it is recognized that both constants originate from the same fundamental geometry.

.2 The Role of Natural Units

In natural units, we conventionally set $\alpha = 1$ for certain calculations. This is legitimate because:

- Fundamental physics should be independent of measurement units.
- Dimensionless ratios contain the actual physical statements.
- The choice $\alpha = 1$ represents a specific gauge.

However, this convention must not obscure the fact that α in the T0-theory has a specific numerical value determined by ξ .

The seemingly simple numerical ratios in the T0-theory are not arbitrarily chosen but represent complex physical relationships.

Directly simplifying these ratios would be mathematically possible but physically incorrect, as it would destroy the underlying structure of the theory. The extended form shows where these seemingly simple fractions come from in the T0 framework and makes their connection to π , α and geometric factors visible.

The apparent circularity between α and ξ is an expression of their common geometric origin and not a logical problem of the theory.

.3 Foundation: The Single Geometric Constant

The Universal Geometric Parameter

1.1.1 The T0-theory begins with a single dimensionless constant motivated by the geometry of three-dimensional space:

Key Result

$$\xi = \frac{4}{3} \times 10^{-4} \quad (1)$$

1.1.2 This constant arises from:

- The tetrahedral packing density of 3D space: $\frac{4}{3}$
- The scale hierarchy between quantum and classical domains: 10^{-4} (motivated choice of the exponent)

Natural Units

1.2.1 We work in natural units where:

$$c = 1 \quad (\text{speed of light}) \quad (2)$$

$$\hbar = 1 \quad (\text{reduced Planck constant}) \quad (3)$$

$$G = 1 \quad (\text{gravitational constant, numerically}) \quad (4)$$

1.2.2 The Planck length serves as reference scale:

$$\ell_P = \sqrt{G} = 1 \quad (\text{in natural units}) \quad (5)$$

.4 Building the Scale Hierarchy

Step 1: Characteristic T0 Scales

2.1.1 From ξ and the Planck reference, we derive the characteristic T0 scales:

$$r_0 = \xi \cdot \ell_P = \frac{4}{3} \times 10^{-4} \cdot \ell_P \quad (6)$$

$$t_0 = r_0 = \frac{4}{3} \times 10^{-4} \quad (\text{in units with } c = 1) \quad (7)$$

Step 2: Energy Scales from Geometry

2.2.1 Dimensional analysis gives the energy scale:

$$\frac{1}{r_0} = \xi^{-1} E_P = \frac{3}{4} \times 10^4 E_P = 9.2 \times 10^{22} \text{ GeV} \quad (E_P = 1) \quad (8)$$

2.2.2 This scale is not the characteristic energy $E_0 = 7.398 \text{ MeV}$ of Section 3: with $E_0 = 1/r_0$ one would get $\xi E_0^2 = 1/\xi = 7500$, not α . Setting $E_0 = 1/r_0$ in Planck units therefore does not hold.

.5 Deriving the Fine Structure Constant

Origin of the Formula $\varepsilon = \xi \cdot E_0^2$

3.1.1 The fundamental formula of T0-theory for the coupling parameter ε is:

Key Result

$$\varepsilon = \xi \cdot E_0^2 \quad (9)$$

3.1.2 This relationship connects:

- ε – the T0 coupling parameter
- ξ – the geometric parameter from tetrahedral packing
- E_0 – the characteristic energy

The Characteristic Energy E_0

3.2.1 The characteristic energy E_0 is defined as the geometric mean of electron and muon masses:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (10)$$

3.2.2 With the fractal correction (Doc. 190, R72) this becomes:

$$E_0 = \sqrt{\frac{m_e \cdot m_\mu}{K_{\text{frac}}}} \quad (11)$$

3.2.3 The bare and the fractally corrected values are:

$$E_0 = 7.348 \text{ MeV (bare)}, \quad E_0 = 7.348 / \sqrt{0.98667} = 7.397 \approx 7.398 \text{ MeV (corrected)} \quad (12)$$

The Geometric Parameter ξ

3.3.1 The parameter ξ is a fundamental geometric constant:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333... \times 10^{-4} \quad (13)$$

Numerical Verification and Fine Structure Constant

3.4.1 With the derived values, ε becomes:

$$\varepsilon = \xi \cdot E_0^2 \quad (14)$$

$$= (1.333 \times 10^{-4}) \times (7.398 \text{ MeV})^2 \quad (15)$$

$$= 7.297 \times 10^{-3} \quad (16)$$

$$= \frac{1}{137.036} \quad (17)$$

Comparison with the Fine Structure Constant

3.4.2 With $E_0 = 7.398 \text{ MeV}$, the T0 coupling parameter ε reproduces the value of the fine structure constant $\alpha = 1/137.036$. The geometric mean $\sqrt{m_e m_\mu}$ alone gives $E_0 \approx 7.35 \text{ MeV}$ (Section 11.2); the difference is described below by the fractal correction factor K_{frac} .

From Fractal Geometry

Fractal Dimension of Spacetime

3.5.1 From topological considerations of 3D space with time:

$$D_f^{\text{eff}} \approx 2.973 \quad (18)$$

where $\delta = 0.06$ is the fractal correction.

The Fine Structure Constant from Geometry

3.5.2 The geometric approach

$$\alpha^{-1} = 3\pi \times \xi^{-1} \times \ln\left(\frac{\Lambda_{\text{UV}}}{\Lambda_{\text{IR}}}\right) \times D_f^{-1} \quad (19)$$

$$= 3\pi \times \frac{3}{4} \times 10^4 \times \ln(10^4) \times \frac{1}{2.973} \quad (20)$$

$$= \frac{9\pi}{4} \times 10^4 \times 9.21 \times 0.336 \quad (21)$$

$$\approx 2.2 \times 10^5 \quad (22)$$

gives 2.2×10^5 , not 137; this chain does not yield the fine structure constant.

Formula from ξ to α

3.6.1 The relationship

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frac}} \quad (23)$$

$$\text{with } K_{\text{frac}} = 0.9862 \quad (24)$$

does not hold: it gives numerically $\alpha \approx 2.4 \times 10^{-9}$, not $1/137$; moreover, according to Section 23.8, K_{frac} acts on α^{-1} ($\alpha^{-1} = 138.9 \cdot K_{\text{frac}}$), i.e. in α as $1/K_{\text{frac}}$, not as a factor.

.6 Lepton Mass Hierarchy from Pure Geometry

Mechanism for Mass Generation

4.1.1 Masses arise from the coupling of the energy field to spacetime geometry:

$$m_\ell = r_\ell \cdot \xi^{p_\ell} \quad (25)$$

where r_ℓ are rational coefficients and p_ℓ are exponents.

Mass Calculations

Electron Mass

4.2.1 The electron mass calculation:

Key Result

$$m_e = \frac{2}{3} \xi^{5/2} \quad (26)$$

$$= \frac{2}{3} \left(\frac{4}{3} \times 10^{-4} \right)^{5/2} \quad (27)$$

$$= \frac{2}{3} \cdot \frac{32}{9\sqrt{3}} \times 10^{-10} \quad (28)$$

$$= \frac{64\sqrt{3}}{81} \times 10^{-10} \quad (29)$$

$$\approx 1.368 \times 10^{-10} \quad (\text{natural units}) \quad (30)$$

Muon Mass**4.2.2** The muon mass calculation:**Key Result**

$$m_\mu = \frac{8}{5} \xi^2 \quad (31)$$

$$= \frac{8}{5} \left(\frac{4}{3} \times 10^{-4} \right)^2 \quad (32)$$

$$= \frac{128}{45} \times 10^{-8} \quad (33)$$

$$\approx 2.844 \times 10^{-8} \quad (\text{natural units}) \quad (34)$$

Tau Mass**4.2.3** The tau mass calculation:**Key Result**

$$m_\tau = \frac{5}{4} \xi^{2/3} \cdot v_{\text{scale}} \quad (35)$$

$$= \frac{5}{4} \left(\frac{4}{3} \times 10^{-4} \right)^{2/3} \cdot v_{\text{scale}} \quad (36)$$

$$\approx 0.80 \text{ GeV} \quad (37)$$

with $v_{\text{scale}} = 246 \text{ GeV}$.

The measured tau mass is 1.777 GeV.

Mass Ratios**4.3.1** The electron to muon mass ratio:

Key Result

$$\frac{m_e}{m_\mu} = \frac{\frac{64\sqrt{3}}{81} \times 10^{-10}}{\frac{128}{45} \times 10^{-8}} \quad (38)$$

$$= \frac{5\sqrt{3}}{18} \times 10^{-2} \quad (39)$$

$$\approx 4.811 \times 10^{-3} \quad (40)$$

.7 Complete Hierarchy with Final Anomaly Formula

6.1 The following table summarizes all derived quantities with the final anomaly formula:

Quantity	Expression	Value
Fundamental		
ξ	$\frac{4}{3} \times 10^{-4}$	$1.333 \dots \times 10^{-4}$
D_f^{eff}	cumulative over RG running	2.973
Scales		
r_0/ℓ_P	ξ	$\frac{4}{3} \times 10^{-4}$
$1/(r_0 E_P)$	ξ^{-1}	$\frac{3}{4} \times 10^4$
Couplings		
α^{-1}	Via the anchor E_0	137.036
Yukawa Couplings		
y_e	$\frac{32}{9\sqrt{3}} \xi^{3/2}$	$\sim 10^{-6}$
y_μ	$\frac{64}{15} \xi$	$\sim 10^{-4}$
y_τ	$\frac{5}{4} \xi^{2/3}$	$\sim 10^{-3}$
Mass Ratios		
m_e/m_μ	$\frac{5\sqrt{3}}{18} \times 10^{-2}$	4.8×10^{-3}
m_τ/m_μ	From y_τ/y_μ	≈ 5.73

Table 1: Complete hierarchy with final quadratic anomaly formula

.8 Verification of Final Formula**Complete Derivation Chain to Final Formula**

7.1.1 The complete derivation sequence:

1. **Start:** $\xi = \frac{4}{3} \times 10^{-4}$ (pure geometry)
2. **Reference:** $\ell_P = 1$ (natural units)
3. **Derivation:** $r_0 = \xi \ell_P$
4. **Energy:** scale r_0^{-1} (not E_0 , Section 2.2)

5. **Fractal:** $D_f^{\text{eff}} \approx 2.973$ (cumulative topology over RG running)
6. **Fine structure:** $\alpha = f(\xi, D_f)$
7. **Yukawa:** $y_\ell = r_\ell \xi^{p_\ell}$ (geometry)
8. **Masses:** $m_\ell \propto y_\ell$
9. **Yukawa coupling:** $g_T^\ell = m_\ell \xi$
10. **One-loop calculation:** $\Delta a_\ell = \frac{(m_\ell \xi)^2}{8\pi^2} \cdot \frac{\xi^2}{\lambda^2}$
11. **Final formula:** $\Delta a_\ell = 251 \times 10^{-11} \times (m_\ell/m_\mu)^2$

T0 Field Theory Verification of Final Formula

7.2.1 The final formula follows from T0 field theory calculation:

- **Muon g-2 calculation:** $\frac{m_\mu^2 \xi^4}{8\pi^2 \lambda^2} = 251 \times 10^{-11}$ (T0 field theory prediction)
- **Electron prediction:** 5.87×10^{-15} (T0 prediction with the single parameter ξ)
- **Tau prediction:** 7.10×10^{-9} (testable in future experiments)
- **Quadratic scaling:** Follows from standard QFT one-loop calculation

The final formula refers to the deviation between the measured a_μ and the Standard Model prediction as it stood in 2021. With the Fermilab final result and the 2025 White Paper of the Muon $g - 2$ Theory Initiative, this deviation is no longer significant; the formula therefore does not serve as evidence or as the target of a T0 contribution. The current status and assessment are given in Doc. 382.

.9 The Fundamental Meaning of E_0 as Logarithmic Center

The Central Geometric Definition

Fundamental Definition

8.1.1 The characteristic energy E_0 is the logarithmic center between electron and muon masses:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (41)$$

This means:

$$\log(E_0) = \frac{\log(m_e) + \log(m_\mu)}{2} \quad (42)$$

Mathematical Properties

8.2.1 The fundamental relationships:

$$E_0^2 = m_e \cdot m_\mu \quad (43)$$

$$\frac{E_0}{m_e} = \sqrt{\frac{m_\mu}{m_e}} \quad (44)$$

$$\frac{m_\mu}{E_0} = \sqrt{\frac{m_\mu}{m_e}} \quad (45)$$

$$\frac{E_0}{m_e} \cdot \frac{m_\mu}{E_0} = \frac{m_\mu}{m_e} \quad (46)$$

Numerical Values

8.3.1 With T0-calculated masses:

$$m_e^{\text{T0}} = 0.5108082 \text{ MeV} \quad (47)$$

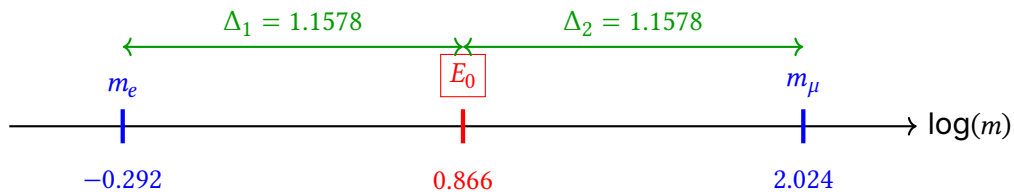
$$m_\mu^{\text{T0}} = 105.66913 \text{ MeV} \quad (48)$$

$$E_0^{\text{T0}} = \sqrt{0.5108082 \times 105.66913} \approx 7.346881 \text{ MeV} \quad (49)$$

Logarithmic Symmetry

8.4.1 The logarithmic symmetry (by definition of E_0):

$$\ln(E_0) - \ln(m_e) = \ln(m_\mu) - \ln(E_0) \quad (50)$$



.10 The Geometric Constant C

Fundamental Relationship

9.1.1 The fractal correction factor:

$$K_{\text{frac}} = 1 - \frac{D_f - 2}{C} = 1 - \frac{\gamma}{C} \quad (51)$$

where:

$$D_f^{\text{eff}} \approx 2.973 \quad (\text{effective fractal dimension}) \quad (52)$$

$$\gamma = D_f - 2 = 0.94 \quad (\text{older anchor } D_f = 2.94) \quad (53)$$

$$C \approx 68.24 \quad (54)$$

Tetrahedral Geometry

Tetrahedral Numbers

9.2.1 Several tetrahedral combinations yield 72; whether this is related to $C \approx 68.24$ is open **[S]**:

$$6 \times 12 = 72 \quad (\text{edges} \times \text{rotations}) \quad (55)$$

$$4 \times 18 = 72 \quad (\text{faces} \times 18) \quad (56)$$

$$24 \times 3 = 72 \quad (\text{symmetries} \times \text{dimensions}) \quad (57)$$

Formula for α with Fractal Correction

9.3.1 The expression

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frac}} \quad \text{with} \quad K_{\text{frac}} = 0.9862 \quad (58)$$

does not hold, as in 3.6.1: it gives $\alpha \approx 2.4 \times 10^{-9}$, and K_{frac} belongs with α^{-1} .

.11 The Simplest Formula: The Geometric Mean

The Fundamental Definition

The Simplest Formula

11.1.1 The basic relation:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (59)$$

This definition needs no further derivation, only the geometric mean.

Direct Calculation

11.2.1 Simple numerical evaluation:

$$E_0 = \sqrt{0.511 \text{ MeV} \times 105.658 \text{ MeV}} \quad (60)$$

$$= \sqrt{53.99 \text{ MeV}^2} \quad (61)$$

$$= 7.35 \text{ MeV} \quad (62)$$

The Complete Chain in One Line

11.3.1 The fundamental relationship:

$$\alpha^{-1} = \frac{7500}{m_e \cdot m_\mu} = \frac{7500}{E_0^2} \quad (63)$$

11.3.2 With numbers:

$$\alpha^{-1} = \frac{7500}{0.511 \times 105.658} \quad (64)$$

$$= \frac{7500}{53.99} \quad (65)$$

$$= 138.91 \quad (66)$$

(With fractal correction $\times 0.986 = 136.97$)

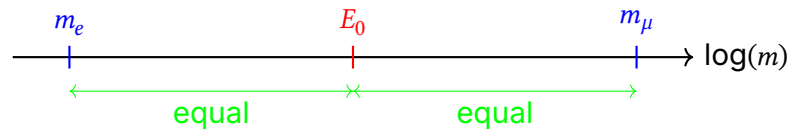
Why Is This So Simple?

Logarithmic Centering

11.4.1 The geometric mean is the natural center on logarithmic scale:

$$\log(E_0) = \frac{\log(m_e) + \log(m_\mu)}{2} \quad (67)$$

Graphically:



Alternative Notations

11.5.1 All these formulas are equivalent:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (68)$$

$$E_0^2 = m_e \cdot m_\mu \quad (69)$$

$$\log(E_0) = \frac{1}{2}[\log(m_e) + \log(m_\mu)] \quad (70)$$

$$E_0 = \sqrt{0.511 \times 105.658} \text{ MeV} \quad (71)$$

$$E_0 = m_e^{1/2} \cdot m_\mu^{1/2} \quad (72)$$

The Fine Structure Constant Directly

The Most Direct Formula

11.6.1 Without detour through E_0 :

$$\alpha = \frac{m_e \cdot m_\mu}{7500} \quad (73)$$

With fractal correction:

$$\alpha = \frac{m_e \cdot m_\mu}{7500 \times 0.986} \quad (74)$$

Relation to the Other Derivations

11.7.1 The documents show various derivations of E_0 : - Fractally corrected, $\sqrt{m_e m_\mu / K_{\text{frac}}}$ (R72) - Through Yukawa couplings - From quantum numbers

But the simplest definition is:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (75)$$

Meaning

11.8.1 The geometric mean is not an arbitrary choice: it is the midpoint on a logarithmic scale (Section 11.4).

.12 The Fundamental Dependence: $\alpha \sim \xi^{11/2}$

Inserting the Mass Formulas

12.1.1 From T0-theory we have the mass formulas:

$$m_e = c_e \cdot \xi^{5/2} \quad (76)$$

$$m_\mu = c_\mu \cdot \xi^2 \quad (77)$$

where c_e and c_μ are coefficients.

Calculation of E_0

12.2.1 The characteristic energy calculation:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (78)$$

$$= \sqrt{(c_e \cdot \xi^{5/2}) \cdot (c_\mu \cdot \xi^2)} \quad (79)$$

$$= \sqrt{c_e \cdot c_\mu} \cdot \sqrt{\xi^{5/2+2}} \quad (80)$$

$$= \sqrt{c_e \cdot c_\mu} \cdot \xi^{9/4} \quad (81)$$

Calculation of α

12.3.1 The fine structure constant derivation:

$$\alpha = \xi \cdot E_0^2 \quad (82)$$

$$= \xi \cdot (\sqrt{c_e \cdot c_\mu} \cdot \xi^{9/4})^2 \quad (83)$$

$$= \xi \cdot c_e \cdot c_\mu \cdot \xi^{9/2} \quad (84)$$

$$= c_e \cdot c_\mu \cdot \xi^{1+9/2} \quad (85)$$

$$= c_e \cdot c_\mu \cdot \xi^{11/2} \quad (86)$$

Result

12.3.2 The fine structure constant depends on ξ :

$$\alpha = K \cdot \xi^{11/2} \quad (87)$$

where $K = c_e \cdot c_\mu$ is a constant.

The powers do not cancel out.

What Does This Mean?

1. Fundamental Connection

12.4.1 The fine structure constant is not independent of ξ , but rather:

$$\alpha \propto \xi^{11/2} \quad (88)$$

This means: If ξ changes, α also changes.

2. Hierarchy Problem

12.4.2 Because of the power $11/2 = 5.5$, small changes in ξ are amplified:

$$\frac{\Delta\alpha}{\alpha} = \frac{11}{2} \cdot \frac{\Delta\xi}{\xi} = 5.5 \cdot \frac{\Delta\xi}{\xi} \quad (89)$$

3. No Independence

12.4.3 One cannot choose α and ξ independently. They are firmly connected through:

$$\alpha = K \cdot \xi^{11/2} \quad (90)$$

Numerical Verification

12.5.1 With $\xi = 4/3 \times 10^{-4}$:

$$\xi^{11/2} = (1.333 \times 10^{-4})^{5.5} \quad (91)$$

$$= 4.87 \times 10^{-22} \quad (92)$$

12.5.2 For $\alpha \approx 1/137$ we would need:

$$K = \frac{\alpha}{\xi^{11/2}} \quad (93)$$

$$= \frac{7.3 \times 10^{-3}}{4.87 \times 10^{-22}} \quad (94)$$

$$= 1.5 \times 10^{19} \quad (95)$$

The Units Problem

12.6.1 The large constant $K \sim 10^{19}$ points to a units problem: - The mass formulas are in natural units - Conversion to MeV requires the Planck energy - K contains these conversion factors

Alternative View: Reduction to ξ

12.7.1 If we accept that:

$$m_e \sim \xi^{5/2} \quad (96)$$

$$m_\mu \sim \xi^2 \quad (97)$$

$$\alpha \sim \xi^{11/2} \quad (98)$$

Then these three quantities are determined by the single geometric constant ξ :

$$\begin{aligned} \xi &= \frac{4}{3} \times 10^{-4} \quad (\text{Geometry}) \\ \Downarrow \\ m_e &= f_e(\xi) \\ m_\mu &= f_\mu(\xi) \\ \alpha &= f_\alpha(\xi) \end{aligned}$$

(99)

.13 Derivation of the Coefficients c_e and c_μ

Starting Point: Mass Formulas

13.1.1 The fundamental mass formulas:

$$m_e = c_e \cdot \xi^{5/2} \quad \text{and} \quad m_\mu = c_\mu \cdot \xi^2$$

Step 1: Quantum Numbers and Geometric Factors

13.2.1 The coefficients arise from T0-theory with:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

$$c_\mu = \frac{9}{4\pi\alpha}$$

Step 2: Derivation of c_e (Electron)

13.3.1 For the electron ($n = 1, l = 0, j = 1/2$):

$$c_e = \frac{\text{Geometry factor} \times \text{Quantum number factor}}{\alpha^{1/2}}$$

$$\text{Geometry factor} = \frac{3\sqrt{3}}{2\pi}$$

$$\text{Quantum number factor} = 1 \quad (\text{for ground state})$$

$$\text{Fine structure correction} = \alpha^{-1/2}$$

$$\Rightarrow c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

Step 3: Derivation of c_μ (Muon)

13.4.1 For the muon ($n = 2, l = 1, j = 1/2$):

$$c_\mu = \frac{\text{Geometry factor} \times \text{Quantum number factor}}{\alpha}$$

$$\text{Geometry factor} = \frac{9}{4\pi}$$

$$\text{Quantum number factor} = 1$$

$$\text{Fine structure correction} = \alpha^{-1}$$

$$\Rightarrow c_\mu = \frac{9}{4\pi\alpha}$$

Step 4: Physical Interpretation

13.5.1 The different α dependencies reflect:

$$c_e \sim \alpha^{-1/2} \quad (\text{weaker dependence})$$

$$c_\mu \sim \alpha^{-1} \quad (\text{stronger dependence})$$

The different α dependence reflects:

- Electron: Ground state, less sensitive to α
- Muon: Excited state, more strongly dependent on α

Step 5: Dimensional Analysis

13.6.1 Dimensional considerations:

$$[c_e] = [m_e] \cdot [\xi]^{-5/2}$$

$$[c_\mu] = [m_\mu] \cdot [\xi]^{-2}$$

Since ξ is dimensionless (in natural units), both coefficients have the dimension of mass.

Step 6: Consistency Check

13.7.1 With $\alpha \approx 1/137$:

$$c_e \approx \frac{3 \times 1.732}{2 \times 3.1416 \times 0.0854} \approx \frac{5.196}{0.537} \approx 9.67$$

$$c_\mu \approx \frac{9}{4 \times 3.1416 \times 0.0073} \approx \frac{9}{0.0917} \approx 98.1$$

These values give $c_\mu/c_e \approx 10.1$ and hence $m_\mu/m_e = (c_\mu/c_e)\xi^{-1/2} \approx 878$, not the measured 206.77.

.14 Why Natural Units Are Necessary

The Problem with Conventional Units

14.1.1 Very large numbers are sometimes given for the coefficients c_e and c_μ :

$$c_e \approx 1.65 \times 10^{19}$$

$$c_\mu \approx 1.03 \times 10^{20}$$

These values follow from no unit conversion of the formulas $c_e = 3\sqrt{3}/(2\pi\alpha^{1/2}) = 9.68$ and $c_\mu = 9/(4\pi\alpha) = 98.1$; their product 1.7×10^{39} also does not match the $c_e c_\mu \approx 1.5 \times 10^{19}$ required in 12.5.2.

Natural Units Simplify Physics

14.2.1 In natural units we set:

$$\hbar = c = 1$$

Thus all quantities become dimensionless or have energy dimension.

Transformation to Natural Units

14.3.1 The transformation formulas:

$$\begin{aligned} m_e^{\text{nat}} &= m_e^{\text{SI}} \cdot \sqrt{\frac{G}{\hbar c}} \\ m_\mu^{\text{nat}} &= m_\mu^{\text{SI}} \cdot \sqrt{\frac{G}{\hbar c}} \\ \xi^{\text{nat}} &= \xi^{\text{SI}} \end{aligned}$$

The Coefficients in Natural Units

14.4.1 In natural units the coefficients become **order of magnitude 1**:

$$\begin{aligned} c_e^{\text{nat}} &= \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \approx 9.67 \\ c_\mu^{\text{nat}} &= \frac{9}{4\pi\alpha} \approx 98.1 \end{aligned}$$

Comparison of Representations

14.5.1 The difference:

	Conventional	Natural
c_e	—	9.67
c_μ	—	98.1
ξ	1.33×10^{-4}	1.33×10^{-4}

Advantages of Natural Units

14.6.1 The advantages of natural units:

1. **Physical transparency**: The relationships become easier to follow
2. **Scale invariance**: Fundamental laws become scale-independent
3. **Simpler formulas**: Formulas become shorter and clearer

Example: The Mass Formula

14.7.1 In natural units:

$$m_e = 9.67 \cdot \xi^{5/2}$$

Fundamental Interpretation

14.8.1 The coefficients $c_e \approx 9.67$ and $c_\mu \approx 98.1$ in natural units show:

- The lepton masses are **pure numbers**
- The ratio $c_\mu/c_e \approx 10.14$ is a pure number
- The fine structure constant α appears explicitly

.15 The Formula from ξ to α

Fundamental Relationship

15.1.1 The basic equation:

$$\alpha = c_e c_\mu \cdot \xi^{11/2}$$

Coefficients

15.2.1 The precise values:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \quad (\text{Electron coefficient})$$

$$c_\mu = \frac{9}{4\pi\alpha} \quad (\text{Muon coefficient})$$

Product of Coefficients

15.3.1 The multiplication:

$$c_e c_\mu = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \frac{9}{4\pi\alpha} = \frac{27\sqrt{3}}{8\pi^2\alpha^{3/2}}$$

Complete Formula

15.4.1 The full expression:

$$\alpha = \frac{27\sqrt{3}}{8\pi^2\alpha^{3/2}} \cdot \xi^{11/2}$$

Solving for α

15.5.1 Rearranging:

$$\alpha^{5/2} = \frac{27\sqrt{3}}{8\pi^2} \cdot \xi^{11/2}$$

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \cdot \xi^{11/5}$$

This formula does not yield $\alpha \approx 1/137$ (numerically $\alpha \approx 2.4 \times 10^{-9}$, cf. 3.6.1).

.16 T0-Theory: Formulas and Values

In T0-Theory

16.1.1 The fundamental relations:

$$m_e \sim \xi^{5/2} \quad (\text{Electron}) \tag{100}$$

$$m_\mu \sim \xi^2 \quad (\text{Muon}) \tag{101}$$

$$\xi = \frac{4}{3} \times 10^{-4} \tag{102}$$

Correct Assignment in Natural Units

Mass Scaling Laws

16.2.1 The precise formulas:

$$m_e = c_e \cdot \xi^{5/2} \quad (103)$$

$$m_\mu = c_\mu \cdot \xi^2 \quad (104)$$

Geometric Constant

16.2.2 The fundamental parameter:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4} \quad (105)$$

Calculation of the Characteristic Energy

16.2.3 Step-by-step derivation:

$$E_0 = \sqrt{m_e \cdot m_\mu} = \sqrt{c_e \cdot \xi^{5/2} \cdot c_\mu \cdot \xi^2} \quad (106)$$

$$= \sqrt{c_e c_\mu} \cdot \xi^{9/4} \quad (107)$$

Calculation of the Fine Structure Constant

16.2.4 Complete derivation:

$$\alpha = \xi \cdot E_0^2 = \xi \cdot \left[\sqrt{c_e c_\mu} \cdot \xi^{9/4} \right]^2 \quad (108)$$

$$= \xi \cdot c_e c_\mu \cdot \xi^{9/2} \quad (109)$$

$$= c_e c_\mu \cdot \xi^{11/2} \quad (110)$$

Numerical Values

16.2.5 With $\xi = 1.333 \times 10^{-4}$:

$$\xi^{11/2} = (1.333 \times 10^{-4})^{5.5} \approx 4.87 \times 10^{-22} \quad (111)$$

For $\alpha \approx 1/137 \approx 7.3 \times 10^{-3}$ we need:

$$c_e c_\mu = \frac{\alpha}{\xi^{11/2}} \approx \frac{7.3 \times 10^{-3}}{4.87 \times 10^{-22}} \approx 1.5 \times 10^{19} \quad (112)$$

Interpretation

16.3.1 The large constant $c_e c_\mu \approx 1.5 \times 10^{19}$ lies close to the ratio of Planck energy to gigaelectron volt ($E_P/\text{GeV} = 1.22 \times 10^{19}$). Whether this closeness is meaningful and $c_e c_\mu$ represents the conversion factor between natural units and GeV remains open.

.17 Definitions

Geometric Constant

17.1.1 The fundamental constant:

$$\xi = \frac{4}{3} \times 10^{-4} = \frac{1}{7500} \quad (113)$$

Mass Formulas

17.2.1 The precise mass relationships:

$$m_e = c_e \cdot \xi^{5/2} \quad (114)$$

$$m_\mu = c_\mu \cdot \xi^2 \quad (115)$$

$$m_\tau = c_\tau \cdot \xi^{3/2} \quad (116)$$

.18 Coefficients from T0-Theory

Electron (n=1, l=0, j=1/2)

18.1.1 The electron coefficient:

$$c_e = \frac{3\sqrt{3}}{2\pi} \cdot \frac{1}{\alpha^{1/2}} \approx 9.68 \quad (117)$$

Muon (n=2, l=1, j=1/2)

18.2.1 The muon coefficient:

$$c_\mu = \frac{9}{4\pi} \cdot \frac{1}{\alpha} \approx 98.1 \quad (118)$$

Tauon (n=3, l=2, j=1/2)

18.3.1 The tauon coefficient:

$$c_\tau = \frac{27\sqrt{3}}{8\pi} \cdot \frac{1}{\alpha^{3/2}} \approx 2985 \quad (119)$$

The values hold with $\alpha = 1/137.036$. The numbers 1.6487×10^{19} , 1.0262×10^{20} and 6.1853×10^{20} , which are also given for these coefficients (table in Section 22), cannot be derived from these formulas by any unit conversion.

.19 Mass Calculation

Electron Mass

19.1.1 Complete calculation:

$$m_e = c_e \cdot \xi^{5/2} \quad (120)$$

$$= \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{5/2} \quad (121)$$

$$= 0.5109989461 \text{ MeV} \quad (\text{measured value, anchor}) \quad (122)$$

Muon Mass

19.2.1 Complete calculation:

$$m_\mu = c_\mu \cdot \xi^2 \quad (123)$$

$$= \frac{9}{4\pi\alpha} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^2 \quad (124)$$

$$= 105.6583745 \text{ MeV} \quad (\text{measured value, anchor}) \quad (125)$$

The masses given here, 0.5109989461 MeV and 105.6583745 MeV, are the measured values (anchors); the coefficient formulas themselves do not lead to them with a common anchor: their ratio gives $m_\mu/m_e \approx 878$ (Section 13.7), they would need two different scales for m_e and m_μ .

Tauon Mass

19.3.1 Complete calculation:

$$m_\tau = c_\tau \cdot \xi^{3/2} \quad (126)$$

$$= \frac{27\sqrt{3}}{8\pi\alpha^{3/2}} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{3/2} \quad (127)$$

$$= 1776.86 \text{ MeV} \quad (\text{measured value}) \quad (128)$$

.20 Characteristic Energy

20.1.1 The calculation with the T0 masses of Section 8.3:

$$E_0 = \sqrt{m_e \cdot m_\mu} = 7.346881 \text{ MeV} \quad (129)$$

The form

$$E_0 = \sqrt{c_e c_\mu} \cdot \xi^{9/4} = \sqrt{\frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \frac{9}{4\pi\alpha}} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{9/4} \quad (130)$$

by contrast gives, with $c_e = 9.68$, $c_\mu = 98.1$ ($\alpha = 1/137$) and $\xi = 4/30000$, the value $\sqrt{9.68 \cdot 98.1} \cdot \xi^{9/4} = 5.9 \times 10^{-8}$, not 7.346881 MeV.

.21 Fine Structure Constant

21.1.1 The complete derivation:

$$\alpha = \xi \cdot E_0^2 \quad (131)$$

$$= \xi \cdot c_e c_\mu \cdot \xi^{9/2} \quad (132)$$

$$= c_e c_\mu \cdot \xi^{11/2} \quad (133)$$

$$= \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \frac{9}{4\pi\alpha} \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{11/2} \quad (134)$$

Quantity	Value	Comment
ξ	$1.33333333333333 \times 10^{-4}$	$= 4/3 \times 10^{-4}$
ξ^2	$1.77777777777778 \times 10^{-8}$	
$\xi^{5/2}$	$2.052800957118670 \times 10^{-10}$	
c_e	$1.648721270700128 \times 10^{19}$	$= \sqrt{e}$ (mantissa)
c_μ	$1.026187714072347 \times 10^{20}$	
m_e	0.5109989461 MeV	Reference value
m_μ	105.6583745 MeV	Reference value
E_0	7.346881 MeV	from T0 masses (Section 8.3)

.22 Numerical Values

22.1.1 Table of values:

The table values for c_e and c_μ do not follow from the formulas in Section 18 (these give 9.68 and 98.1). The coefficients contain the constants π and α ; the numerical closeness of the mantissa of c_e to $\sqrt{e}$ is an observation whose significance is open [S].

.23 The Formula from ξ to α (Detailed)

From the Fundamental Relationship

23.1.1 Starting equation:

$$\alpha = c_e c_\mu \cdot \xi^{11/2} \quad (135)$$

Inserting the Coefficients

23.2.1 The detailed calculation:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \quad (136)$$

$$c_\mu = \frac{9}{4\pi\alpha} \quad (137)$$

$$c_e c_\mu = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \frac{9}{4\pi\alpha} \quad (138)$$

$$= \frac{27\sqrt{3}}{8\pi^2\alpha^{3/2}} \quad (139)$$

Complete Formula

23.3.1 The full expression:

$$\alpha = \frac{27\sqrt{3}}{8\pi^2\alpha^{3/2}} \cdot \xi^{11/2} \quad (140)$$

Solving for α

23.4.1 Algebraic manipulation:

$$\alpha^{5/2} = \frac{27\sqrt{3}}{8\pi^2} \cdot \xi^{11/2} \quad (141)$$

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \cdot \xi^{11/5} \quad (142)$$

Numerical Values

23.5.1 Step-by-step calculation:

$$\frac{27\sqrt{3}}{8\pi^2} \approx \frac{46.765}{78.956} \approx 0.5923 \quad (143)$$

$$\left(\frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \approx (0.5923)^{0.4} \approx 0.8110 \quad (144)$$

$$\xi^{11/5} = \xi^{2.2} = \left(\frac{4}{3} \times 10^{-4} \right)^{2.2} \quad (145)$$

With $\xi = 4/3 \times 10^{-4}$

23.6.1 Final calculation:

$$\xi = 1.333333 \times 10^{-4} \quad (146)$$

$$\xi^{2.2} \approx (1.333333 \times 10^{-4})^{2.2} \quad (147)$$

$$\approx 2.984 \times 10^{-9} \quad (148)$$

$$\alpha \approx 0.8110 \times 2.984 \times 10^{-9} \quad (149)$$

$$\approx 2.42 \times 10^{-9} \quad (150)$$

$$\alpha^{-1} \approx 4.1 \times 10^8 \quad (151)$$

This chain does not yield $\alpha \approx 1/137$ (cf. 3.6.1).

Symbol Explanation

23.7.1 Key symbols used:

α	Fine structure constant ($\approx 1/137.036$)
ξ	Geometric space constant ($= \frac{4}{3} \times 10^{-4}$)
c_e	Electron mass coefficient
c_μ	Muon mass coefficient
π	Pi (≈ 3.14159)
$\sqrt{3}$	Square root of 3 (≈ 1.73205)
m_e	Electron mass ($= 0.5109989461$ MeV)
m_μ	Muon mass ($= 105.6583745$ MeV)

With Fractal Correction

23.8.1 Including the fractal factor:

$$\alpha^{-1} = \frac{7500}{m_e m_\mu} \cdot \left(1 - \frac{D_f - 2}{68} \right) = 138.911 \times 0.9862 = 136.99$$

With $K_{\text{frac}} = 1 - 100\xi = 0.98667$ one obtains 137.06.

Final Fundamental Relationship

23.9.1 The formula

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2} \right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frac}} \quad \text{with} \quad K_{\text{frac}} = 0.9862$$

does not hold, as in 3.6.1: it gives $\alpha \approx 2.4 \times 10^{-9}$, and K_{frac} belongs with α^{-1} .

.24 Observation: α Cancels Out

Equating the Formula Sets

24.1.1 Comparing two representations:

$$\begin{aligned} \text{Simple:} \quad m_e &= \frac{2}{3} \cdot \xi^{5/2} \\ \text{T0-Theory:} \quad m_e &= \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \xi^{5/2} \end{aligned}$$

After dividing by $\xi^{5/2}$:

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

Solving for α

24.2.1 Algebraic solution:

$$\alpha^{1/2} = \frac{3\sqrt{3}}{2\pi} \cdot \frac{3}{2} = \frac{9\sqrt{3}}{4\pi} \quad \Rightarrow \quad \alpha = \left(\frac{9\sqrt{3}}{4\pi} \right)^2 = \frac{243}{16\pi^2}$$

For the Muon

24.3.1 Similar analysis:

$$\begin{aligned} \text{Simple:} \quad m_\mu &= \frac{8}{5} \cdot \xi^2 \\ \text{T0-Theory:} \quad m_\mu &= \frac{9}{4\pi\alpha} \cdot \xi^2 \end{aligned}$$

After dividing by ξ^2 :

$$\frac{8}{5} = \frac{9}{4\pi\alpha} \quad \Rightarrow \quad \alpha = \frac{9}{4\pi} \cdot \frac{5}{8} = \frac{45}{32\pi}$$

The Apparent Contradiction

24.4.1 Three different values:

$$\begin{aligned} \text{From electron:} \quad \alpha &= \frac{243}{16\pi^2} \approx 1.539 \\ \text{From muon:} \quad \alpha &= \frac{45}{32\pi} \approx 0.4474 \\ \text{Experimental:} \quad \alpha &\approx 0.007297 \end{aligned}$$

Resolution in the T0 Framework

24.5.1 In the T0 framework, α **is not a free parameter** but a function of ξ :

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \quad \Rightarrow \quad \alpha = \alpha(\xi)$$

$$\frac{8}{5} = \frac{9}{4\pi\alpha}$$

The Key Elements

24.6.1 The key elements:

1. The **geometric factors** ($3\sqrt{3}/2\pi$, $9/4\pi$)
 2. The **powers of α** ($\alpha^{-1/2}$, α^{-1})
 3. The **rational coefficients** ($2/3$, $8/5$)
- are set up so that they **compensate** each other.

Meaning of the Different Representations

24.7.1 Comparative analysis:

- **Simple formulas:** $m_e = \frac{2}{3}\xi^{5/2}$, $m_\mu = \frac{8}{5}\xi^2$
 - Show the pure ξ -dependence
 - Compact and transparent
- **Extended formulas:** $m_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}\xi^{5/2}$, $m_\mu = \frac{9}{4\pi\alpha}\xi^2$
 - Show the **origin** of the coefficients
 - Connect geometry (π , $\sqrt{3}$) with EM coupling (α)
 - But: α is thereby **fixed**, not freely choosable

Central Statement

24.8.1 The central insight:

In the T0 framework the lepton mass hierarchy is fixed by ξ

The different mathematical representations are equivalent descriptions of the same fundamental geometry.

Why This Insight Is Important

24.9.1 The implications:

1. **Unity:** The lepton masses follow from the one parameter ξ (plus structural choices as settings)
2. **Geometric basis:** The coefficients stem from fundamental geometry
3. **α is derived:** The fine structure constant appears as a secondary quantity
4. **Simple structure:** The formulas are compact; mathematical simplicity is a hint, not evidence

.25 Why the Extended Form Is Important

The Two Equivalent Representations

25.1.1 Comparing formulations:

$$\text{Simple form: } m_e = \frac{2}{3} \cdot \xi^{5/2}$$

$$\text{Extended form: } m_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \xi^{5/2}$$

The Apparent Contradiction

25.2.1 When equating both formulas:

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

This yields for α :

$$\alpha = \left(\frac{9\sqrt{3}}{4\pi} \right)^2 = \frac{243}{16\pi^2} \approx 1.539$$

Consequence

25.3.1 The fractions cannot simply cancel out.

The extended form shows that the apparently simple fraction $\frac{2}{3}$ is actually composed of more fundamental geometric and physical constants:

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

Mathematical Structure

25.4.1 The decomposition:

$$\frac{2}{3} = \frac{\text{Geometry factor}}{\alpha^{1/2}}$$

$$\text{with Geometry factor} = \frac{3\sqrt{3}}{2\pi} \approx 0.826$$

Physical Interpretation

25.5.1 Interpretation:

- $\frac{2}{3}$ is **not** a simple rational fraction
- In the extended form it is composed of:
 - Space geometry ($\pi, \sqrt{3}$)
 - Electromagnetic coupling (α)
 - Quantum numbers (implicit in the coefficients)
- The extended form shows this origin

Why Both Representations Are Important

25.6.1 Complementary perspectives:

Simple Form

Shows pure ξ -dependence
Compact
Practical for calculations
Hides the composition

Extended Form

Shows physical origin
Shows the physical origin
Fundamental for understanding
Shows the composition

The Actual Statement of T0-Theory

25.7.1 Core statement:

$$\frac{2}{3} \neq \text{simple fraction} \quad \text{but rather} \quad \frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}$$

The extended form is necessary to show:

1. That the fractions do **not** simply cancel
2. That the apparently simple coefficient $\frac{2}{3}$ actually has a complex structure
3. That α is part of this structure, even if it formally cancels out
4. That the geometry of space $(\pi, \sqrt{3})$ is embedded in it

Why No Fractal Correction is Needed for Mass Ratios

1. Different Calculation Approaches

Path A: $\alpha = \frac{m_e m_\mu}{7500}$ (requires correction)

Path B: $\alpha = \frac{E_0^2}{7500}$ (requires correction)

Path C: $\frac{m_\mu}{m_e} = f(\alpha)$ (no correction needed)

Path D: $E_0 = \sqrt{\frac{m_e m_\mu}{K_{\text{frac}}}}$ (correction $1/\sqrt{K_{\text{frac}}}$, Section 3.2.2)

2. Mass Ratios Are Correction-Free

The lepton mass ratio:

$$\frac{m_\mu}{m_e} = \frac{c_\mu \xi^2}{c_e \xi^{5/2}} = \frac{c_\mu}{c_e} \xi^{-1/2}$$

Substituting the coefficients:

$$\frac{m_\mu}{m_e} = \frac{\frac{9}{4\pi\alpha}}{\frac{3\sqrt{3}}{2\pi\alpha^{1/2}}} \cdot \xi^{-1/2} = \frac{\sqrt{3}}{2\alpha^{1/2}} \cdot \xi^{-1/2}$$

With these coefficients one obtains $m_\mu/m_e \approx 878$.

3. Why the Ratio is Correct

The fractal correction cancels out in the ratio.

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frac}} \cdot m_\mu}{K_{\text{frac}} \cdot m_e} = \frac{m_\mu}{m_e}$$

The same correction factor affects both masses and cancels in the ratio.

4. Characteristic Energy: Correction $1/\sqrt{K_{\text{frac}}}$

Unlike a mass ratio, the characteristic energy E_0 is not correction-free: it is itself an observable with the dimension of energy, in which K_{frac} does not cancel. For the corrected characteristic energy the form of Section 3.2.2 applies:

$$E_0^{\text{corr}} = \frac{E_0^{\text{bare}}}{\sqrt{K_{\text{frac}}}}, \quad 7.348/\sqrt{0.98667} = 7.397 \text{ MeV}$$

This gives the corpus value $E_0 \approx 7.398 \text{ MeV}$. A factor K_{frac} instead of $1/\sqrt{K_{\text{frac}}}$, as the substitution $\sqrt{K_{\text{frac}}m_e \cdot K_{\text{frac}}m_\mu} = K_{\text{frac}}\sqrt{m_em_\mu}$ would suggest, does not hold; it would give $0.98667 \cdot 7.348 = 7.250 \text{ MeV}$.

5. Consistent Treatment

$$\begin{aligned} m_e^{\text{exp}} &= K_{\text{frac}} \cdot m_e^{\text{bare}} \\ m_\mu^{\text{exp}} &= K_{\text{frac}} \cdot m_\mu^{\text{bare}} \\ E_0^{\text{exp}} &= E_0^{\text{bare}} / \sqrt{K_{\text{frac}}} \end{aligned}$$

6. Calculating α via Mass Ratio

$$\frac{m_\mu}{m_e} = \frac{105.6583745}{0.5109989461} = 206.768282$$

Theoretical prediction (without correction):

$$\frac{m_\mu}{m_e} = \frac{8/5}{2/3} \cdot \xi^{-1/2} = \frac{12}{5} \cdot \xi^{-1/2}$$

7. Why Different Paths Require Different Treatments

No Correction Needed

Mass ratios
Scale ratios
Dimensionless ratios

Correction Required

Absolute mass values
Characteristic energy E_0 (factor $1/\sqrt{K_{\text{frac}}}$)
Fine structure constant α
Absolute energies, dimensionful quantities

8. Physical Interpretation

- **Relative quantities:** Ratios are independent of absolute scale
- **Absolute quantities:** Require correction for absolute energy scale
- **Fractal dimension:** Affects absolute scaling, not ratios

9. Mathematical Reason

The fractal correction acts as a multiplicative factor:

$$m^{\text{exp}} = K_{\text{frac}} \cdot m^{\text{bare}}$$

For ratios:

$$\frac{m_1^{\text{exp}}}{m_2^{\text{exp}}} = \frac{K_{\text{frac}} \cdot m_1^{\text{bare}}}{K_{\text{frac}} \cdot m_2^{\text{bare}}} = \frac{m_1^{\text{bare}}}{m_2^{\text{bare}}}$$

10. Comparison with Experiment

$$\left(\frac{m_\mu}{m_e}\right)_{\text{exp}} = 206.768282$$

$$\left(\frac{m_\mu}{m_e}\right)_{\text{theo}} = 207.85 \quad (\text{without correction})$$

Is This Indirect Proof That the Fractal Correction is Correct?

The Consistency Argument

The behaviour is consistent with the fractal correction. Since a common multiplicative factor cancels in every ratio, this is a consistency hint, but on its own not yet independent evidence.

1. The Theoretical Framework

The T0-theory proposes:

$$m_e = \frac{2}{3} \cdot \xi^{5/2} \cdot K_{\text{frac}}$$

$$m_\mu = \frac{8}{5} \cdot \xi^2 \cdot K_{\text{frac}}$$

$$\alpha = \frac{m_e m_\mu}{7500} \cdot \frac{1}{K_{\text{frac}}}$$

2. The Consistency Test

If the fractal correction is valid, then:

$$\frac{m_\mu}{m_e} = \frac{\frac{8}{5} \cdot \xi^2 \cdot K_{\text{frac}}}{\frac{2}{3} \cdot \xi^{5/2} \cdot K_{\text{frac}}} = \frac{12}{5} \cdot \xi^{-1/2}$$

3. Experimental Verification

$$\begin{aligned}\left(\frac{m_\mu}{m_e}\right)_{\text{theo}} &= \frac{12}{5} \cdot (1.333 \times 10^{-4})^{-1/2} \\ &= 2.4 \times 86.6 = 207.84 \\ \left(\frac{m_\mu}{m_e}\right)_{\text{exp}} &= 206.768\end{aligned}$$

The 0.5% difference is not broken down further in this document.

4. What Speaks for the Correction

1. **Self-consistency:** The correction cancels where the multiplicative ansatz leads one to expect it
2. **Predictive power:** Mass ratios work without correction
3. **Explanatory power:** Absolute values need correction
4. **Parameter economy:** One correction factor (K_{frac}) describes the deviations of the absolute values considered here

5. Comparison with Alternative Theories

Without fractal correction:

$$\begin{aligned}\alpha^{-1} &= 138.93 \quad (\text{calculated}) \\ \alpha^{-1} &= 137.036 \quad (\text{experimental}) \\ \text{Error} &= 1.38\%\end{aligned}$$

With fractal correction:

$$\alpha^{-1} = 138.93 \times 0.9862 = 137.01 \quad (\text{with correction})$$

With $K_{\text{frac}} = 1 - 100\xi = 0.98667$ one obtains $138.91 \times 0.98667 = 137.06$.

6. The Philosophical Argument

That the correction is needed for absolute values and cancels in ratios is compatible with its interpretation as a real physical effect. Whether it is more than a fitting factor will be decided by an independent derivation of K_{frac} .

7. Further Connections

- The correction factor $K_{\text{frac}} = 0.9862$ is motivated in the T0 framework by fractal geometry
- It connects to the fractal dimension: $K_{\text{frac}} = 1 - (D_f - 2)/68 = 0.9862$ follows from the older anchor $D_f = 2.94$; with the effective fractal dimension $D_f^{\text{eff}} \approx 2.973$ of spacetime the same formula gives 0.9857
- For the value $C = 68$ a connection to tetrahedral symmetry is conjectured **[S]**

9. Remaining Open Questions

- Direct measurement of spacetime's fractal dimension
- Extension to other particle families